EDA=> Process of examining, understanding, cleaning and visualizing a dataset before using it for further analysis or building an ML model.

Exploring data to understand what it is inside, find problems, discover patterns, and learn what data is telling.

EDA helps to discover=> patterns, missing data, outliers, relationships, data quality, data distribution.

Use:

- Understand the data (rows, columns, which columns contains number or which contains text, what column represents)
- To find missing and duplicate values
- To understand data distribution or identify outliers
- To discover relationships between columns
- To prepare data

Purpose:

- What data do i have?
- Is my data correct and complete?
- How is my data distributed?
- What patterns exist in the data?
- What should i do next?

Application:

- Market analysis and customer segmentation
- Risk assessment in finance and insurance
- Quality control in manufacturing
- Healthcare data analysis and disease prediction
- Recommendation system and product optimization

Types of EDA: Variables is a column in dataset

- **Univariate =>** Analyze one variable at a time. **Common Techniques:** mean, median, mode, min and max, standard deviation, frequency count, histogram, box plots, box chunks, **Purpose:** To understand the characteristics and distribution of a single variable. **Real-time example:** Finding the average age of students in a university.
- **Bivariate=>** Analyze two variables to understand their relationships. **Common Techniques:** correlation, cross-tabulation, scatter plot, line graph, group comparison, comparing avg between categories. **Purpose:** To understand whether two variables are related or different. **Real-time example:** Checking whether older students get higher marks?

- **Multivariate=>** Analyze two or more variables together. **Common Techniques:** Correlation matrix, pair plots, multiple variable visualization, principal component analysis (PCA), spatial analysis, cluster analysis**. Purpose:** To understand more complex relations involving multiple variables. **Real-time example:** Company analyzes age, experience, education, and salary together to understand employee salary.

STEPS for EDA:

- **STEP 1: Understand the problem**: Before analyzing data, decide what we want to learn from it or what problem we are solving.
    - It helps focus analysis on the questions that matter instead of calculating unnecessary statistics.
- **STEP 2: Load the dataset=>** Common format: csv, excel, Json, Db
- **STEP 3: Understand the dataset=>** After loading the dataset, find out how much data and what it contains.
    - Check: The number of rows, the number of columns, the column names, A sample of the data, Whether the values look correct.
    - It helps to understand the size and general structure of dataset before making changes.
- **STEP 4: Check Data Type**=> Incorrect data types can cause errors in calculations and make analysis unreliable.
- **STEP 5: Clean the data**=> Real-world datasets are not always perfect. They may contain:
    - Duplicate rows, Incorrect values, Extra space, Inconsistent category names, Missing values, Numbers stored as text.
- **STEP 6: Find and handling missing values=>** Remove rows, fill missing values, use suitable replacement.
    - To prevent missing data from producing misleading results.
- **STEP 7: Univariate Analyze=>** analyze each column separately.
    - **For numerical columns:** calculate: Mean. Median. Minimum. Maximum. Standard deviation. Distribution of values.
    - For **categorical columns:** count how frequently each category appears.
- **STEP 8: Bivariate or multivariate analyze=>** investigate whether two or more variables are related.
- **STEP 9: Detect Outliers=>** Common methods include: Minimum and maximum checks. Box plots. Interquartile range (IQR). Z-scores.
    - Outliers may indicate incorrect data, unusual events, or genuine extreme values. Investigate them before deciding whether to correct, retain, or remove them.
    - Apply techniques like capping or transformation if necessary.
- **STEP 10: Visualize the data=>** Histogram, bar chart, boxplot, scatter plot, line chart

Understand the data -> Load data -> Inspect data -> Check data Types-> Clean data -> Handle missing data -> Analyze variable -> Explore relations -> Detect outliers -> visualize variable -> Summarize findings.

Feature Engineering=> The process of creating new features(columns), modifying existing features, or transforming data to make it more useful to analyze.

Why do we use it?

- Create useful information
- Improve data understanding
- Improve ml model
- Simplify data
- Identify relationships

Types:

1. **Creating new features=>** Creating a new column using existing columns. **E.g.=>** math English science subjects marks are columns, we add total marks, percentage and average.
2. **Feature Transformation=>** Changing the format or values of existing features to make it more useful.
3. **Encoding categorical data=>** Contains group or labels rather than numerical measurements. Encoding converts categorical values into numbers, so algorithms can process them.
   E.g.:
   a. **One-hot Encoding**=> Creates a separate column for each category and uses 0 and 1 to represent it.
   b. **Label Encoding**=> Assign each category a unique number.
   c. **Ordinal Encoding**=> Assigns numbers to categories according to their natural order or ranking.
   d. **Frequency Encoding**=> Replaces each category with the number of times it appears in the dataset, or its frequency.

Technique	What does it do?	Example
One-Hot Encoding	Creates separate 0/1 columns	Male → 1, Female → 0
Label Encoding	Assigns a unique number	Male → 0, Female → 1
Ordinal Encoding	Assigns numbers according to order	Low → 1, Medium → 2, High → 3
Frequency Encoding	Assigns numbers based on occurrence count	Southampton → 3

4. **Feature Scaling=>** Changes the range of numerical values so that features can be compared or used more efficiently by certain ml algo. Common technique=>

a. min-max Scaling/ Normalization=> Transform 0-1 range
b. Standardization/Z-score => Based on mean and standard deviation

Feature Selection=> Choose the most useful existing features. Usually, remove or exclude less useful columns. Changes the range of numerical values to make features comparable.

Hypothesis Testing=> Help us to determine whether there is enough statistical evidence to support a claim or whether the observed result could simple be due to chance.

Used=> Test Assumption-> Check whether an assumption is supported by data, make decisions, compare groups, identify relationships, avoid misleading conclusion.

Types of Hypotheses:

1. **Null Hypothesis (H_0) =>** There is no effect no difference or no relationships being tested.
2. **Alternative Hypothesis (H_1 or H_α)**=> State that there is an effect difference or relationships.

STEPS:

- **STEP-1: Define the problem=>**decide what you want to investigate.
- **STEP-2: Formulate Hypothesis=>**Create two hypotheses.
- **STEP-3: Select a significance level=>**The significance level is represented by alpha (α). It is the threshold used to decide whether statistical evidence is strong enough to reject the null hypothesis. A commonly used significance level is: **α = 0.05 (5%)**
- **STEP-4: Choose a statistical Testing=>** Select an appropriate statistical test based on data and research question.

Statistical Test	Purpose	Example
T-test	Compares the means of two groups	Comparing marks of two student groups
ANOVA	Compares the means of three or more groups	Comparing marks across three teaching methods
Chi-Square Test	Checks the association between categorical variables	Gender and course preference
Pearson Correlation Test	Tests the strength of a linear relationship between two numerical variables	Study hours and marks
Z-test	Tests hypotheses about means or proportions under suitable conditions	Comparing a sample mean with a known population mean

- **STEP-5: Calculate the P-value=>** A **p-value** tells us how surprising the observed results (or more extreme results) would be if the null hypothesis were true. Compare the p-value with the

significance level. If p-value <=0.05 -> reject the null hypothesis, if p-value >0.05-> Fail to reject the null hypothesis.

- **STEP-6: Draw a conclusion=>**Explains whether there is sufficient evidence to reject the null hypothesis.

Final Steps for Performing EDA:

Understanding the problem-> Loading the data -> Understanding the dataset -> Checking data types -> Cleaning the data -> Finding and handling missing values -> Univariate analysis -> Bivariate analysis -> Multivariate analysis -> Outlier detection -> Feature engineering -> Hypothesis testing -> Data visualization -> Drawing conclusions.

Node.js Libraries:

1. **Simple Statistics:** Used to perform statistical calculations such as mean, median, and mode.
2. **Danfo.js:** Used for loading, cleaning, and analyzing datasets.
3. **Plotly.js:** Used for creating data visualizations, charts, and graphs.

Examples:

- https://www.kaggle.com/code/mdnahidurrahmankh/complete-exploratory-data-analysis-eda#2.-Load-the-Data
- https://medium.com/@sanjayskumar4010/comprehensive-guide-to-exploratory-data-analysis-eda-with-python-f6e4ad672113
- https://www.kaggle.com/code/imoore/intro-to-exploratory-data-analysis-eda-in-python
- https://www.kaggle.com/code/senakoak/exploratory-data-analysis-examples/notebook
- https://medium.com/@sanjayskumar4010/comprehensive-guide-to-exploratory-data-analysis-eda-with-python-f6e4ad672113
- https://www.kaggle.com/code/imoore/intro-to-exploratory-data-analysis-eda-in-python
- https://www.kaggle.com/code/senakoak/exploratory-data-analysis-examples/notebook